



8214 - Mining JWST data for hidden asteroid gems

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

While the smallest bodies in Earth vicinity are abundant, they are challenging to observe due to their faintness. Yet, they hold the keys to major endeavors ranging from studying meteorite sources (and thus planetary-defense efforts) to understanding the collisional and dynamical evolution of asteroidal bodies through the solar system history (2023-2032 Decadal Survey Q4.1). Fortunately, JWST's unique infrared capabilities covering the emission peaks of asteroids located up to 10 au, can be combined with synthetic-tracking techniques to detect main-belt asteroids (MBAs) as small as 10 m. Recently, 138 of such decameter MBAs were detected serendipitously crossing MIRI/F1500W's field of view (FoV) during exoplanet observations of the TRAPPIST-1 system. These observations enabled for the first time to sample a size regime associated with meteorite parent bodies, commonly transported from MBAs to near-earth objects (NEOs). They revealed a break in the sampled size-frequency distribution (SFD) around 100 m—as expected for a population driven by collisional cascade with the lowest material strength at this diameter. Here, we propose a

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Legacy Archival Research program to mine 10x more data with FoVs sampling a wide range of inclinations. The larger data set will yield O(600) decameter MBAs and thus provide (1) strong constraints on the SFD's slopes, (2) an inclination-dependence study of the SFDs, and (3) a color-dependence of the SFDs (using data in various MIRI filters). Associations of decameter MBAs to asteroid families, informed by the corresponding inclinations and colors, will be used for studies of the meteorite source regions (in particular of H, L, CM, or CI classes).

OBSERVING DESCRIPTION